

Math Class 9th Federal

Board Chapter 08 Geometry

of Straight Lines- Full Test

[SLO: M-09-B-03]: Find the gradient of a straight line when coordinates of two points are given.	Summative
[SLO: M-09-B-04]: Find the equation of a straight line in the form $y = mx + c$	Summative
[SLO: M-09-B-05]: Find the gradient of parallel and perpendicular lines.	Summative
[SLO: M-09-B-08]: Derive equation of a straight line in slope-intercept form, point-slope form, two-point form, intercepts form, symmetric form, normal form.	Summative
[SLO: M-09-B-09]: Show that a linear equation in two variables represents a straight line and reduce the general form of the equation of a straight line to the other standard forms.	Summative

Angle Between Lines	[SLO: M-09-B-10]: Find the angle between two coplanar intersecting straight lines.	Summative
	[SLO: M-09-B-11]: Find the equation of the family of lines passing through the point of intersection of two given lines.	Summative
	[SLO: M-09-B-12]: Calculate angles of the triangle when the slopes of the sides are given.	Summative

$$m = \frac{8-0}{0-4} = -2$$

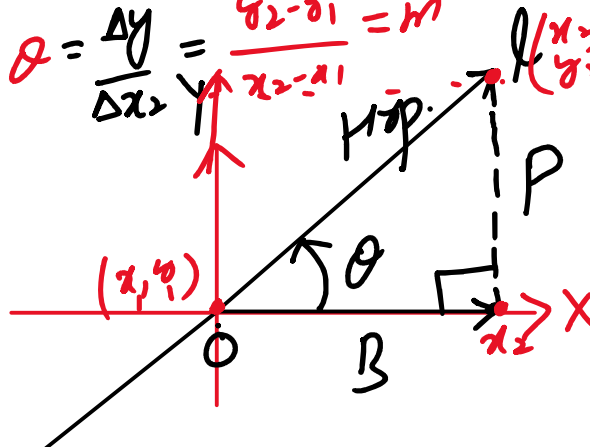
7. Slope of a line passing through two points $(4,0)$ and $(0,-8)$ is:	-2	0.5	2	0
8. The angle between the given lines is: $y = \frac{3}{4}x - 5$ $y = \frac{3}{4}x + 5$ is:	0°	45°	90°	180°

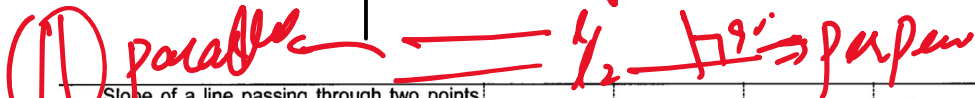
Slope = Gradient

$$m = \frac{0 - (-8)}{4 - 0} = \frac{8}{4} = 2$$

$$m = \tan \theta = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = m$$

$$T = \frac{P}{B}$$





7.	Slope of a line passing through two points (4, 0) and (0, -8) is: خط (4, 0) اور (0, -8) سے گزرنے والی لائن کی سلوپ:	-2	0.5	2	0
8.	The angle between the given lines is: $y = \frac{3}{4}x - 5$ $y = \frac{3}{4}x + 5$ is: دو لائنوں کا باہمی زاویہ کیا ہے؟	0°	45°	90°	180°

→ standard form of straight line

$y = mx + c$

$\frac{3}{4}$
 m slope / gradient

c y-intercept

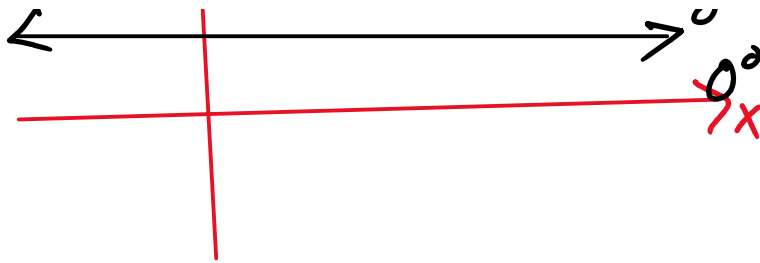
$m_1 = m_2 \rightarrow$ lines are parallel.

Angle is

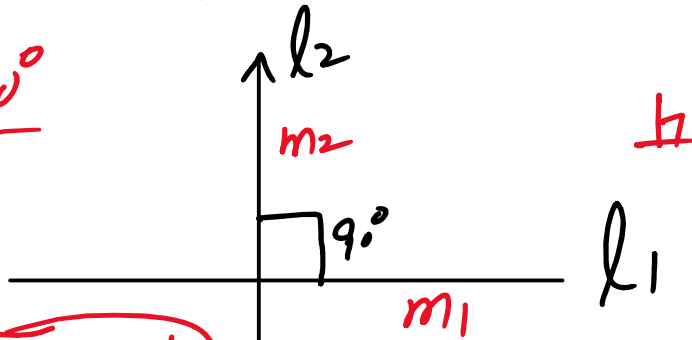
0°

m_1 l_1
 m_2 l_2
 $\Rightarrow m_1 = m_2$





$\theta = 90^\circ$



$m_1 \times m_2 = -1$

If ² lines are $\perp$ each other,
then product of slopes = -1

$l_1 \rightarrow m_1 = 2$

$l_2 \perp l_1, m_2 = ?$

$m_2 = -\frac{1}{2}$

$2 \times -\frac{1}{2} = -1$

$m_1 \times m_2 = -1$

$2 \times m_2 = -1$

$m_2 = -\frac{1}{2}$

7. Slope of a line passing through two points (3,5) and (9,11) is:

$m = \frac{11-5}{9-3} = \frac{6}{6} = 1$

$\frac{8}{3}, \frac{4}{3}, \frac{6}{6}, \frac{8}{4}$

7. Slope of a line passing through two points (3,5) and (9,11) is: $\frac{8}{3}$ $\frac{4}{3}$ $\frac{6}{6}$ $\frac{8}{4}$

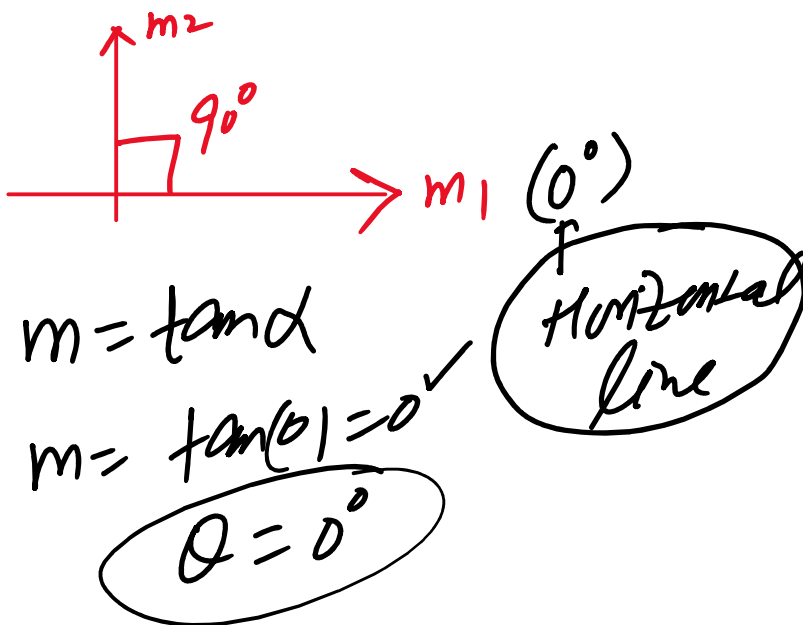
At what angle do the given lines intersect each other? $y = \frac{2}{3}x - 4$ $y = \frac{-3}{2}x + 4$

3. 0° 45° 90° 180°

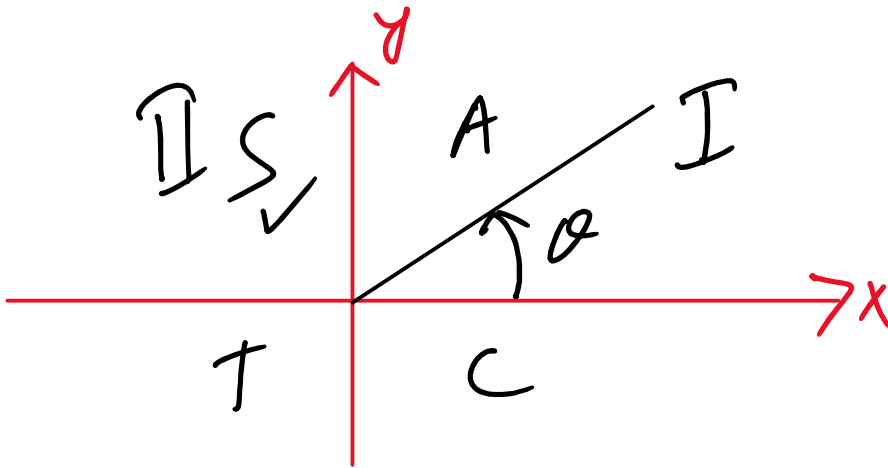
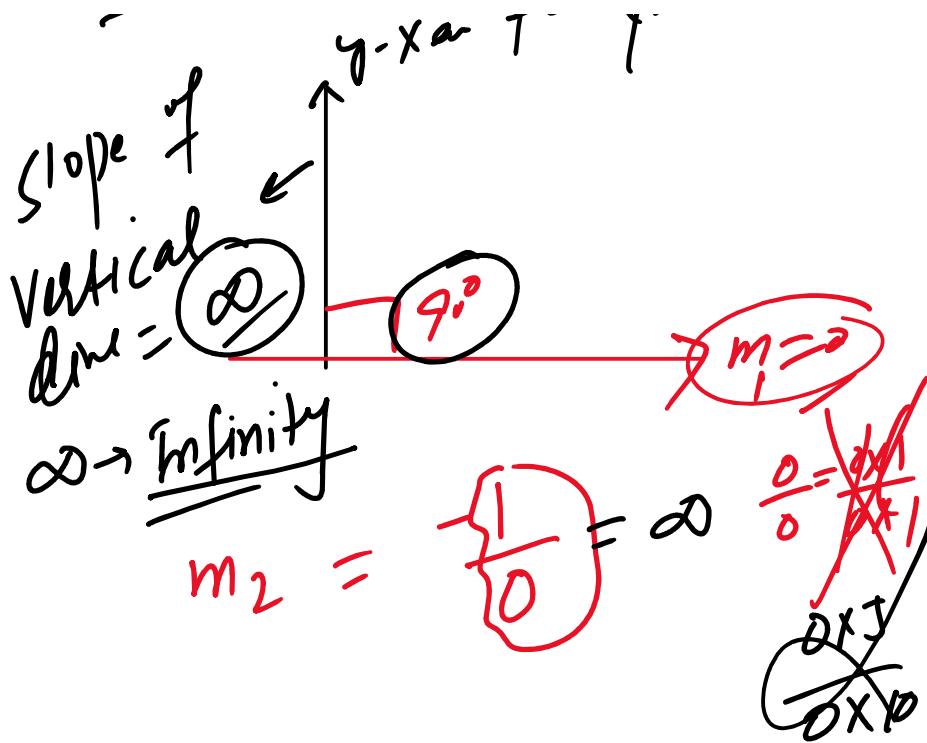
Handwritten notes: $9-3=6$, $11-5=6$, $\frac{6}{6}=1$, $-\frac{3}{2} \times \frac{2}{3} = -1$, $1 + (-1) = 0 \Rightarrow 90^\circ$

1. Encircle the correct option in the following.
- (i) Slope of two lines are 0 and ∞ . What is angle between the lines? (a) 0° (b) 60° (c) 90° (d) 180°
- (ii) One angle of right triangle ABC is determined by the line joining A(1, 2) and B(2, 3). Find the third angle. (a) 30° (b) 45° (c) 60° (d) 80°
- (iii) Slope of (-1, 6) and (1, y) is 3. What is the value of y? (a) 10 (b) 11 (c) -12 (d) 12
- (iv) The line $5x - ky - 3 = 0$ passes through (1, 2). What is the value of k? (a) 1 (b) -1 (c) -2 (d) 2

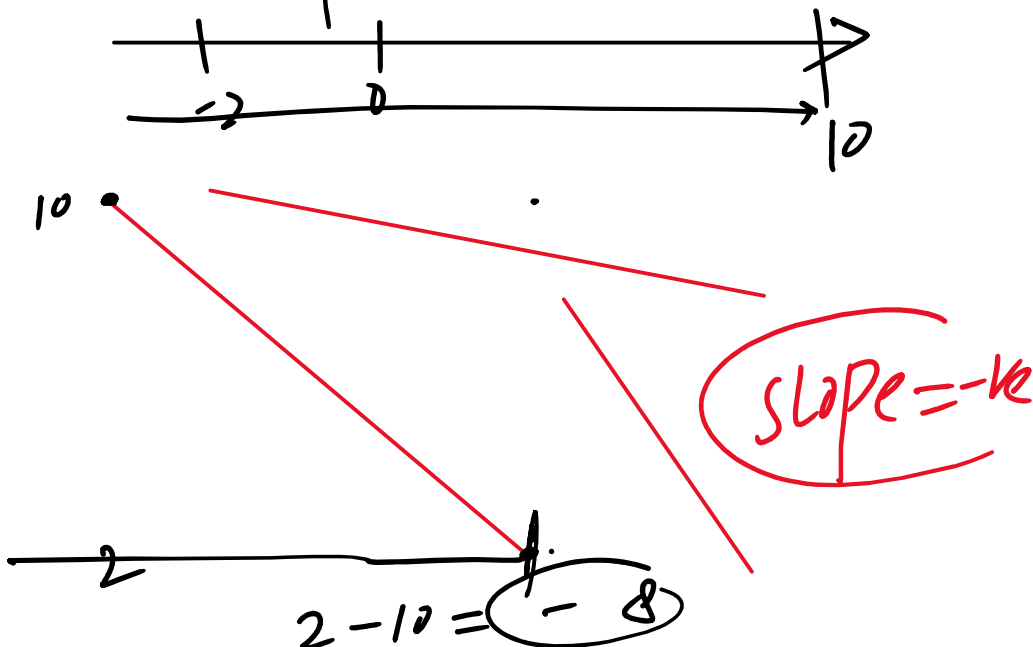
0° m_1 $m_2 \Rightarrow m_1 = m_2$



Vertical line is always $\parallel$ to y-axis

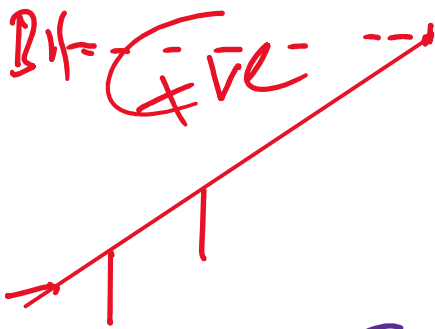
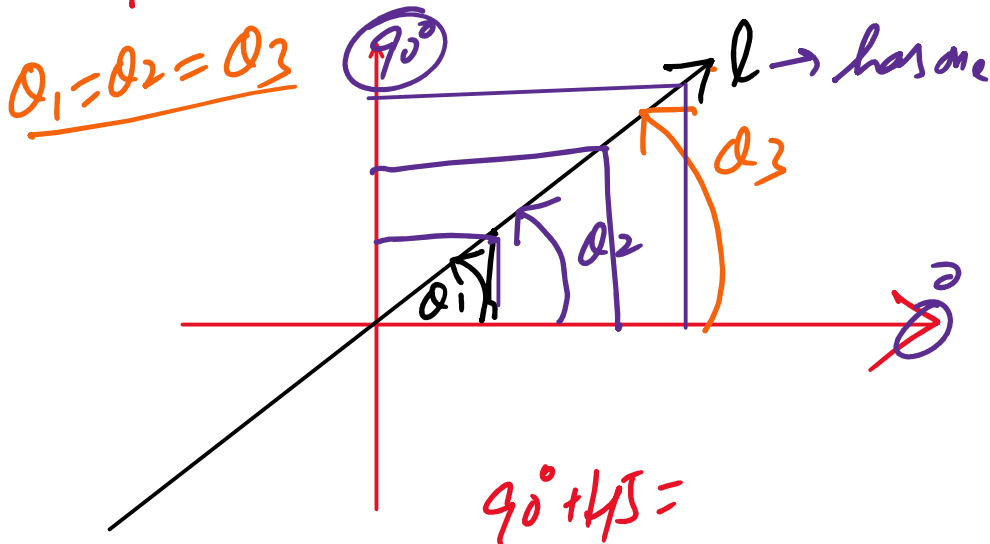


Left to Right



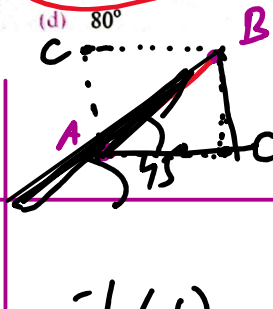
$$2 - 10 = \textcircled{-8}$$

Diff = $\vec{r}_2 - \vec{r}_1$

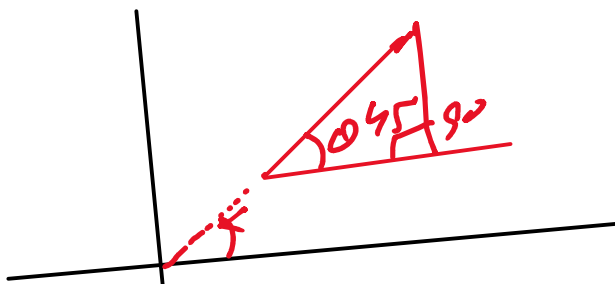



- (ii) One angle of right triangle ABC is determined by the line joining A(1, 2) and B(2, 3). Find the third angle.
- (a) 30° (b) 45° (c) 60° (d) 80°

$$m = \frac{3-2}{2-1} = \frac{1}{1} = \textcircled{1}$$



$$m = \tan \alpha = d = \tan^{-1}(1)$$



- (iii) Slope of $(-1, 6)$ and $(1, y)$ is 3. What is the value of y?
- (a) 10 (b) 11 (c) -12 (d) 12

(iii) Slope of $(-1, 6)$ and $(1, y)$ is 3. What is the value of y ?

(a) 10

(b) 11

(c) -12

(d) 12

$$m = 3 = \frac{y - 6}{1 - (-1)} = \frac{y - 6}{2} \Rightarrow 3$$

$$y - 6 = 6$$

$$y = 12$$

(iv) The line $5x - ky - 3 = 0$ passes through $(1, 2)$. What is the value of k ?

(a) 1

(b) -1

(c) -2

(d) 2

(v) The line $5x - 6 = 0$ represents a line:

(a) parallel to x-axis

(b) parallel to y-axis

(c) passing through origin

(d) touching both axis

$$5(1) - k(2) - 3 = 0$$

$$5 - 2k - 3 = 0$$

$$-2k = -2 \Rightarrow k = 1$$

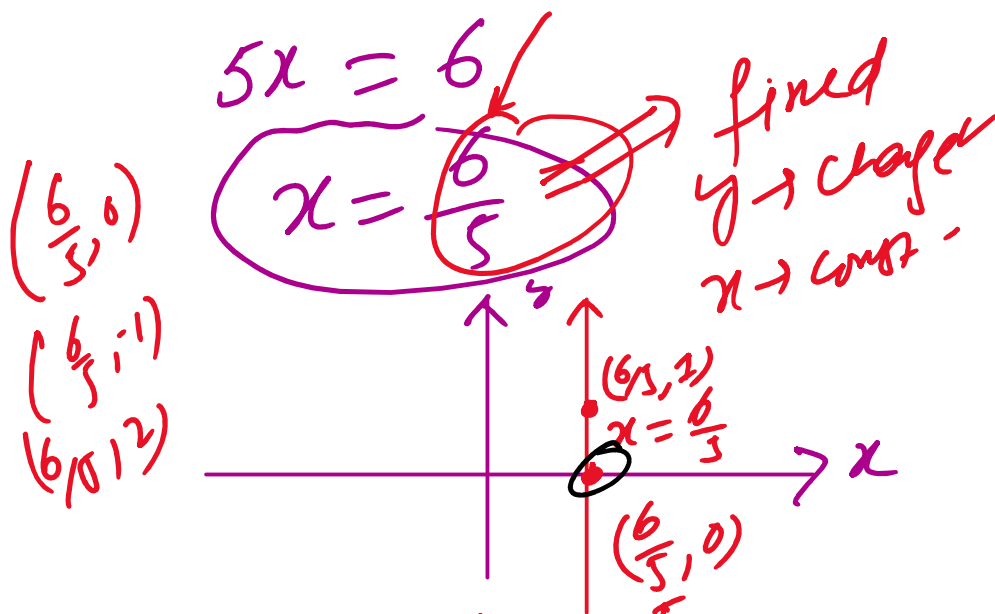
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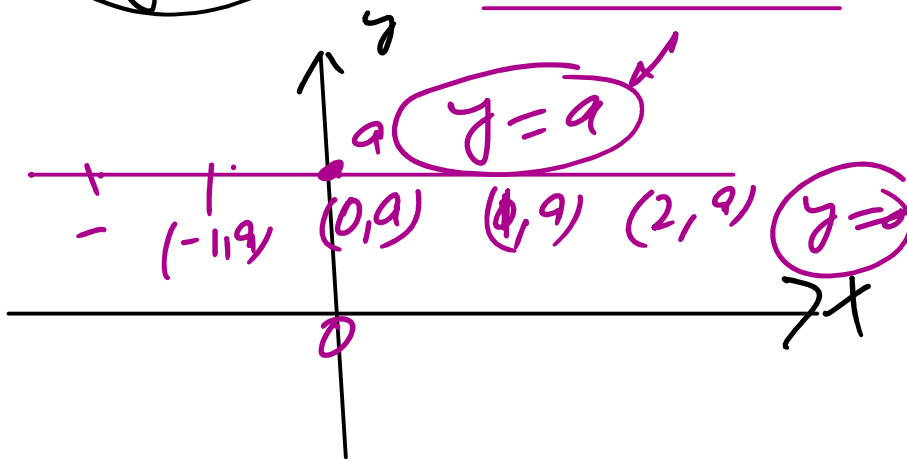
(d) touching both axis



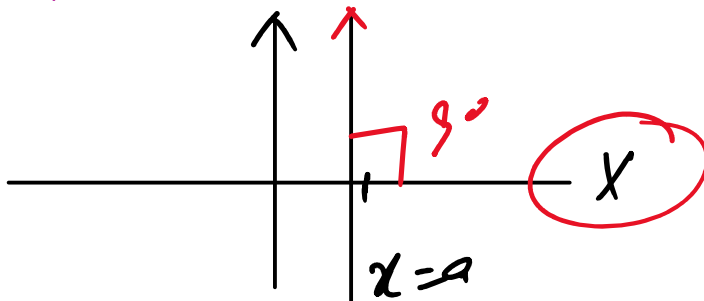
$$x=0 \quad \left| \quad \left(\frac{0}{5}, 0 \right) \right.$$

$x=a \Rightarrow$ parallel to y-axis

$$y=a$$



$x=a$ is perp to
 a) x-axis, b) y-axis
 c) both axes d) origin



(vi)

Line $y=0$ represents:

(a) x-axis

(b) y-axis

(c) a plane

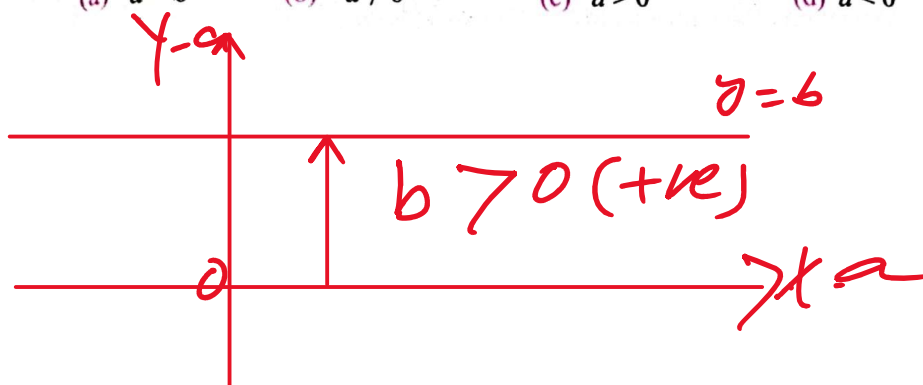
(d) a point

$$x=0 \rightarrow y\text{-axis}$$

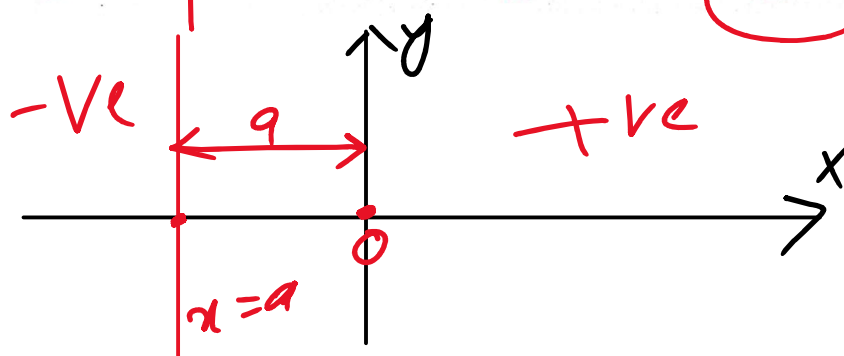
(X)

$x = a$ $y = b$

- (vii) The line $y = b$, is above the x-axis, if
 (a) $b = 0$ (b) $b < 0$ (c) $b > 0$ (d) $b \neq 0$
- (viii) The line $x = a$, is left to the y-axis, if
 (a) $a = 0$ (b) $a \neq 0$ (c) $a > 0$ (d) $a < 0$



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- (ix) Slope of a line l is -4 . What is slope of a line perpendicular to l ?
 (a) $\frac{1}{4}$ (b) $-\frac{1}{4}$ (c) 4 (d) -4
- (x) Which of the following line has slope $\frac{2}{3}$?
 (a) $2x + 3y = 0$ (b) $2x - 3y = 2$ (c) $3x - 2y = 1$ (d) $3x + 2y = 3$

$l \Rightarrow m_1 = -4, m_2 = \frac{1}{4}$

$m_1 \times m_2 = -1$



(ix) Slope of a line l is -4 . What is slope of a line perpendicular to l ?

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$m = \frac{2}{3}$

$y = mx + c$

$ax + by + c = 0 \rightarrow$ general form

$by = -ax - c$

$\rightarrow y = -\frac{a}{b}x - \frac{c}{b}$

$m = -\frac{\text{coefficient of } x}{\text{coefficient of } y}$

$-\frac{a}{b} = -\frac{2}{3} = \frac{2}{3}$

$2x - 3y = 2$

$-3y = -2x + 2$

$y = mx + c$
Slope m , Intercept c

$y = \frac{2}{3}x + \frac{2}{3}$
 m, c

(xi) The line $y = 5x - 3$ is written in the form:

- (a) point-slope (b) two-intercept (c) slope-intercept (d) two-point

(xii) x-intercept of the line $x + y = 5$ is:

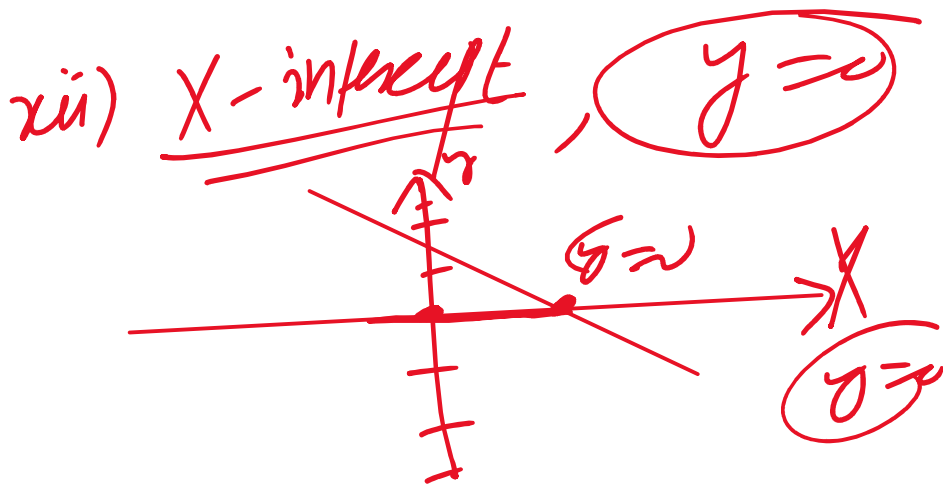
- (a) 1 (b) -1 (c) 5 (d) -5

(xiii) A line intersects both axis at $(2, 0)$ and $(0, 7)$ respectively. Its y-intercept is:

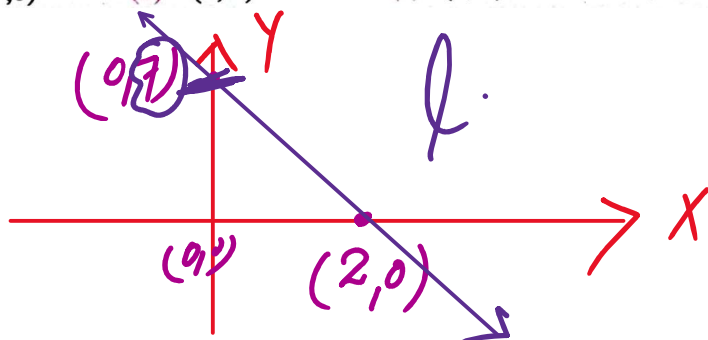
- (a) -2 (b) 2 (c) -7 (d) 7

(xiv) Point of intersection of lines $9x - 7y = 0$ and $8x - 11y = 0$, is:

- (a) $(0, 0)$ (b) $(0, 7)$ (c) $(9, 0)$ (d) $(7, 9)$

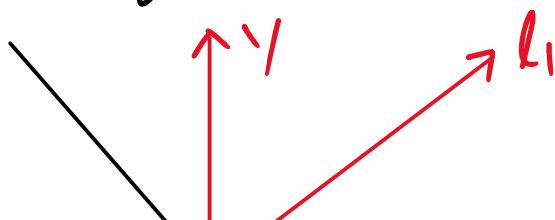


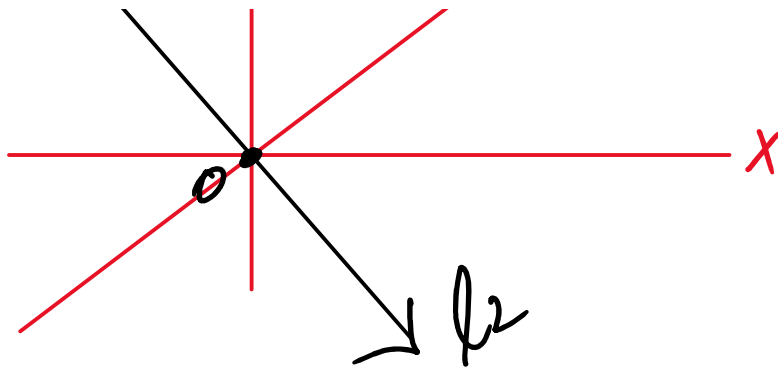
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 (a) $(0, 0)$ (b) $(0, 7)$ (c) $(9, 0)$ (d) $(7, 9)$



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- (xiv) Point of intersection of lines $9x - 7y = 0$ and $8x - 11y = 0$, is:
 (a) $(0, 0)$ (b) $(0, 7)$ (c) $(9, 0)$ (d) $(7, 9)$

$(0, 0)$
 $ax + by + c = 0$
 $\Rightarrow c = 0$
 $\Downarrow$ it always passes through origin





Model Paper ✓✓

11, 8, 9, 4, 6

Find the equation of a line through the point $(1, 3)$ and the point of intersection of the lines $2x - 3y + 4 = 0$ and $4x + y - 1 = 0$.

Sol: Given point $P(1, 3)$

$$\begin{aligned} x &= -\frac{1}{13} \\ y &= \frac{9}{7} \end{aligned}$$

$$2x - 3y + 4 = 0 \rightarrow \textcircled{1}$$

$$4x + y - 1 = 0 \rightarrow \textcircled{2}$$

Eq. $\textcircled{1} \times -2$: $-4x + 6y - 8 = 0$ Adding

$\textcircled{2}$: $4x + y - 1 = 0$

$$7y - 9 = 0$$

$$7y = 9 \Rightarrow \boxed{y = \frac{9}{7}}$$

put value of y in Eqn (1), right

$$2x - 3\left(\frac{9}{7}\right) + 4 = 0$$

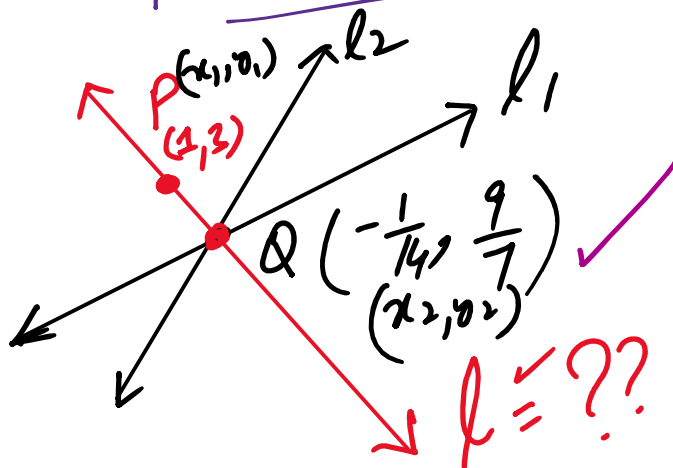
$$2x - 3\left(\frac{9}{7}\right) + 4 = 0$$

$$\times 7 \Rightarrow 14x - 27 + 28 = 0$$

$$14x + 1 = 0$$

$$14x = -1$$

$$\Rightarrow x = -\frac{1}{14}$$



By using Two-points form of Eqn. of straight line

$$\begin{matrix} (x_1, y_1) \\ P(1, 3) \\ Q(-\frac{1}{14}, \frac{9}{7}) \\ (x_2, y_2) \end{matrix}$$

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

$$y - 3 = \left(\frac{\frac{9}{7} - 3}{-\frac{1}{14} - 1} \right) (x - 1)$$

$$\left(\frac{-\frac{1}{14} - 1}{\frac{1}{14} - 1} \right)$$

$$y - 3 = \frac{8}{5}(x - 1)$$

$$\times 5 \Rightarrow 5y - 15 = 8x - 8$$

$$8x - 5y + 15 - 8 = 0$$

$$8x - 5y + 7 = 0$$



Find the gradient of the line passing through the points $A(2, 3)$ and $B(6, 11)$.

Solⁿ As we know that (x_1, y_1) (x_2, y_2)

$$\text{Gradient of a line} = m = \frac{y_2 - y_1}{x_2 - x_1}$$

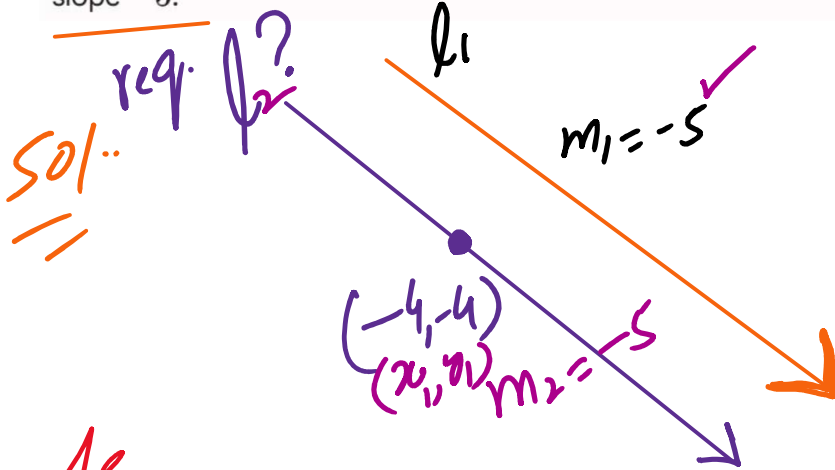
$$m = \frac{11 - 3}{6 - 2}$$

$$\Rightarrow m = \frac{8}{4} = 2$$

$$\rightarrow \boxed{m=2}$$

Thus, Gradient of the required line is 2

Find the equation of the straight line passing through $(-4, -4)$ and parallel to the line with slope -5 .



As ~~given~~ slope of given line $= m_1 = -5$

As the req. line is $\parallel$ to the given line, hence

slope of the req. line $= m_2 = m_1$

$$\rightarrow \boxed{m_1 = m_2 = m = -5}$$

$$\Rightarrow \underline{m_1 = m_2 = m}$$

Also, the eq. line passes through

$P(-4, 4)$, By using
 ~~(x_1, y_1)~~

Point-slope form, ~~use~~
 $y - y_1 = m(x - x_1)$

$$y - (-4) = -5(x - (-4))$$

$$y + 4 = -5x - 20$$

$$5x + y + (4 + 20) = 0$$

$$5x + y + 24 = 0$$

$(-4, 4)$
 $-20 - 4 + 25$

$$m = -5$$

is the very

$(x_1, y_1) (x_2, y_2)$

Find the gradient of the line passing through $(-4, -3)$ and $(2, 9)$. Also write the equation of this line

Find the gradient of the line passing through $(-4, -3)$ and $(2, 9)$. Also write the equation of this line.

Sol.
2

$$m = \frac{9 - (-3)}{2 - (-4)} = \frac{12}{6} = 2$$

By using point-slope form

$$y - y_1 = m(x - x_1)$$

$$y - (-3) = 2(x - (-4))$$

$$y + 3 = 2x + 8$$

$$2x - y + 8 - 3 = 0$$

$y = mx + c$

$$2x - y + 5 = 0$$

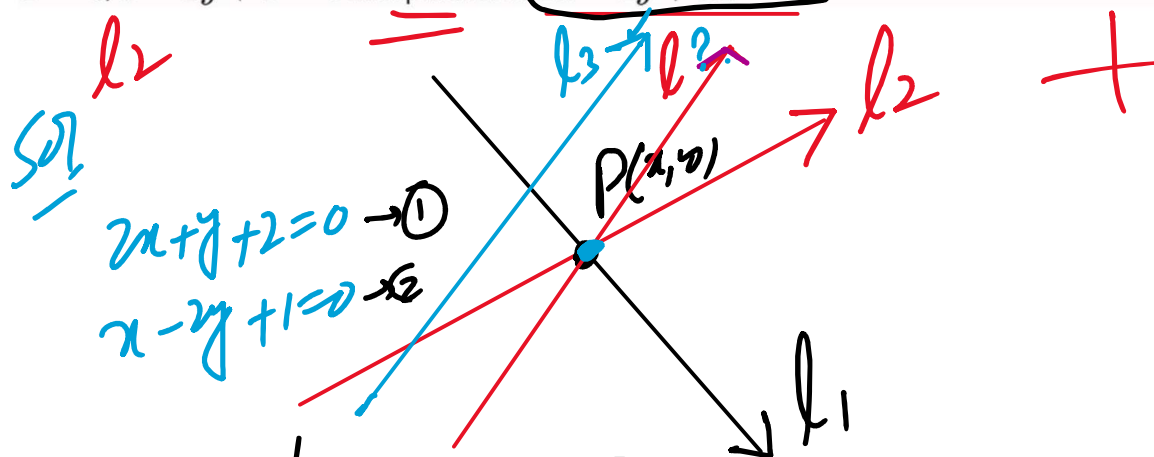
$$\Rightarrow y = 2x + 5$$

$-\frac{1}{2}$

$-\frac{2}{1}$

$\downarrow 4$

Find the equation of the line passing through the point of intersection of the lines $2x + y + 2 = 0$, $x - 2y + 1 = 0$ and parallel to $2x - 2y + 11 = 0$.



$$\begin{aligned} \textcircled{1} \times 2: 4x + 2y + 4 &= 0 \\ \textcircled{2}: x - 2y + 1 &= 0 \end{aligned} \quad \begin{array}{c} + \\ - \end{array}$$

$$\hline 5x + 5 = 0 \Rightarrow x = -1$$

$$\textcircled{1} \Rightarrow 2(-1) + y + 2 = 0$$

$$\begin{aligned} -2 + y + 2 &= 0 \\ \Rightarrow y &= 0 \end{aligned}$$

$$P(-1, 0)$$

The slope of the line $2x - 2y + 11 = 0$

$$m = -\frac{2}{-2} = 1$$

$$m = 1$$

As the req. line is $\parallel$ to the given line $2x - 2y + 11 = 0$

✓ ✓ ✓ ✓ ✓
Slope of the req. line = $m = 1$

By using point-slope for

$$y - y_1 = m(x - x_1)$$

$$y - 0 = 1(x - (-1))$$

$$y = x + 1$$

Ans is the req. line ✓

⑥ ✓ Find the equation of straight-line having y -intercept = 4 and parallel to the line with slope $\frac{1}{2}$. ✓

Sol ✓ The slope of req. line = $\frac{1}{2}$ (due to 11 lines)

$$\Rightarrow m = \frac{1}{2} \checkmark$$

$$\text{also } y\text{-intercept} = -4 = c \checkmark$$

Also, y -intercept $= -4 = c$

By using slope-intercept form

$$y = mx + c$$

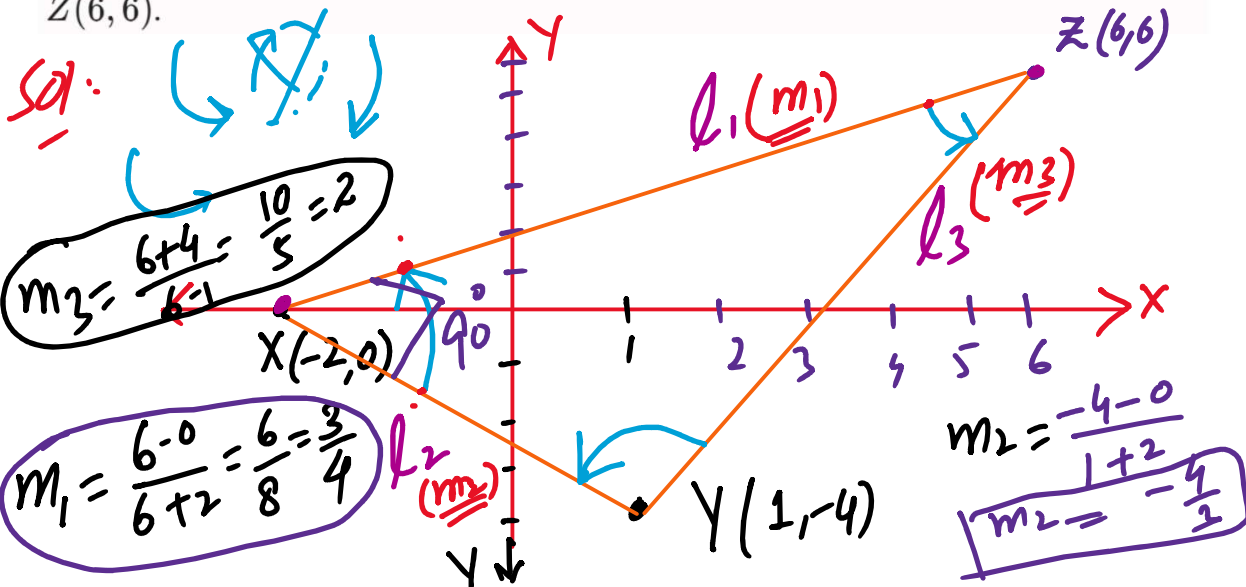
$$y = \frac{1}{2}x - 4$$

OR $\times 2 \rightarrow 2y = x - 8$

$$x - 2y - 8 = 0$$

Q1 $\hookrightarrow$ Counter-clockwise

Find the interior angles of triangle XYZ whose vertices are $X(-2, 0)$, $Y(1, -4)$, and $Z(6, 6)$.



$\therefore$ have 2 lines in $\triangle XYZ$.

We have 3 lines in $\triangle XYZ$.

$\overline{XY}$ line has a slope m_2

$\overline{XZ}$ " " " " " " m_1

$\overline{YZ}$ " " " " " " m_3

As we know that formula for angle b/w two lines.

$$\tan X = \frac{m_1 - m_2}{1 + m_1 m_2} \rightarrow (A)$$

$$\tan Y = \frac{m_2 - m_3}{1 + m_2 m_3} \rightarrow (B)$$

$$Z = 180 - X - Y \rightarrow (C)$$

Ex:

$$(A) \Rightarrow \tan X = \frac{\frac{3}{4} + \frac{4}{3}}{1 + \left(\frac{3}{4}\right)\left(\frac{4}{3}\right)}$$

$$\tan X = \frac{\frac{25}{12}}{0} = \infty$$

∞

∞

$$\frac{1}{0}$$

$$X = \tan^{-1}\left(\frac{25}{0}\right) = 90^\circ$$

$$\boxed{X = 90^\circ}$$

$$\text{Eqn (1)} \Rightarrow \tan Y = \frac{-\frac{4}{3} - 2}{1 + (-\frac{4}{3})(2)}$$

$$\Rightarrow Y = \tan^{-1}(\downarrow)$$

$$\boxed{Y = 63.435^\circ}$$

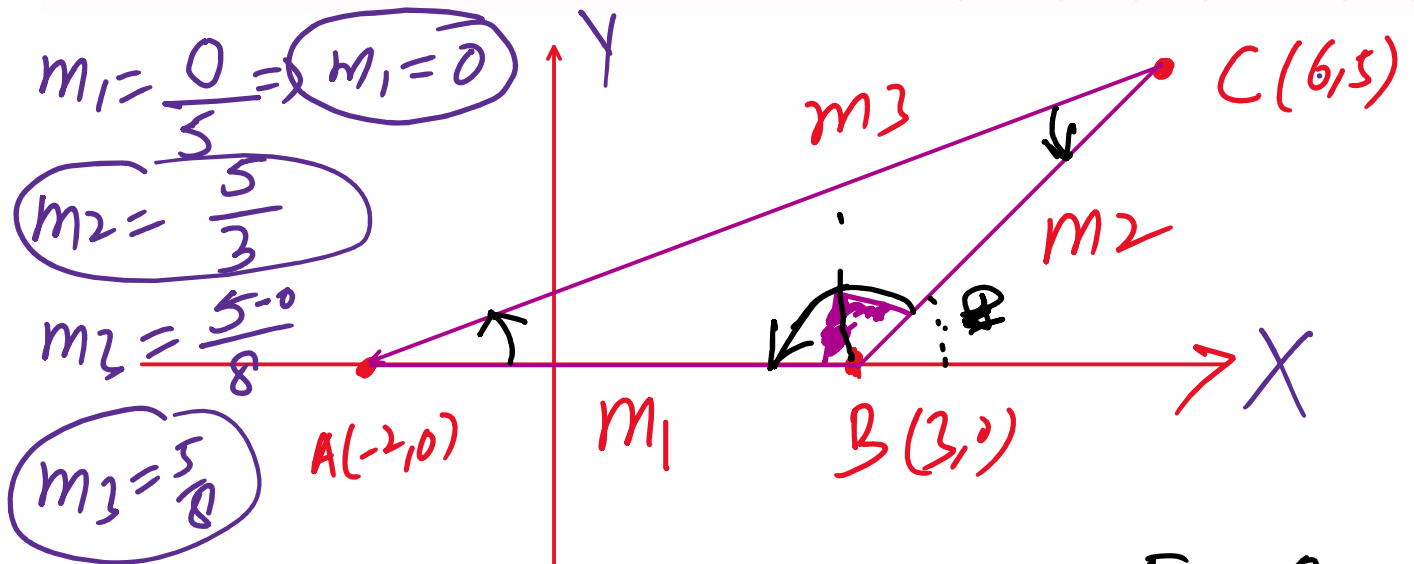
$$\text{And Eqn (C)} \Rightarrow Z = 180 - 90^\circ - 63.435^\circ$$

$$\boxed{Z = 26.565^\circ}$$

$$\text{Thus } \boxed{V = 90^\circ, V = 63.435^\circ, Z = 26.565^\circ}$$

Thus $\boxed{X = 90^\circ, Y = 63.43^\circ, Z = 26.57^\circ}$

Find the interior angles of the triangle whose vertices are $A(-2, 0)$, $B(3, 0)$, and $C(6, 5)$.

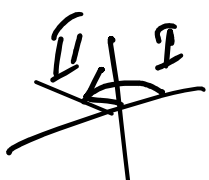


$$\tan A = \frac{m_3 - m_1}{1 + m_1 m_3} = \frac{5 - 0}{1 + 0}$$

$$\tan A = \frac{5}{8} \Rightarrow A = \tan^{-1}\left(\frac{5}{8}\right)$$

$$\Rightarrow \boxed{A = 32.005^\circ} \checkmark$$

$$\tan B = \frac{m_1 - m_2}{1 + m_1 m_2} = \frac{0 - \frac{5}{3}}{1 + 0}$$



$-\frac{5}{3}$ / $\text{Rin II}^{\text{nd}}$ quad



$$\tan B = -\frac{5}{3} \quad (B \text{ in II Quad})$$

$$B = \tan^{-1}\left(\frac{5}{3}\right) \rightarrow \text{Reference angle}$$

$$B = 59.036$$

As $\tan B$ lies in 2nd quad

$$B = 180 - 59.036$$

$$B = 120.964^\circ$$

$$\text{As } A + B + C = 180^\circ$$

$$C = 180^\circ - 32.05^\circ - 120.965^\circ$$

$$C = 27.031^\circ$$

$$0 \leq \alpha \leq 180^\circ$$

Convert the equation $3x + 4y - 12 = 0$ into the given forms:

a. Slope intercept

b. Two intercepts ✓

c. Symmetric

d. Normal

↳ General form

$$ax + by + c = 0$$

a) $y = mx + c$ → Required

Sol:
2

$$3x + 4y - 12 = 0$$

$$\Rightarrow 4y = -3x + 12$$

$$\Rightarrow y = -\frac{3}{4}x + 3$$

This is the req. form of slope-intercept

Here $m = -\frac{3}{4}$ $c = 3$

Here, $\boxed{m = -\frac{3}{4}}$, $\boxed{C = 3}$

Convert the equation $3x + 4y - 12 = 0$ into the given forms:

- a. Slope intercept
- b. Two intercepts
- c. Symmetric
- d. Normal

⑥ Two intercept form is given

$$\boxed{\frac{x}{a} + \frac{y}{b} = 1}$$

where $a = x\text{-intercept}$, $b = y\text{-intercept}$

$$3x + 4y - 12 = 0$$

$$\Rightarrow 3x + 4y = 12$$

$\div$ by 12 on both sides, we get

$$\frac{3x}{12} + \frac{4y}{12} = \frac{12}{12}$$

$$\Rightarrow \boxed{\frac{x}{4} + \frac{y}{3} = 1}$$

$$\Rightarrow \left| \left| \frac{x}{4} + \frac{y}{3} = 1 \right| \right|$$

is the req. two-intercept form

where $a = x\text{-intercept} = 4$

$b = y\text{-intercept} = 3$ ✓

$$ax + by + c = 0,$$

c)

$$3x + 4y - 12 = 0$$

The symmetric form

$$\boxed{\frac{x-x_1}{\cos \alpha} = \frac{y-y_1}{\sin \alpha}}$$

is required
one.

→ A

★ Slope of given straight line is

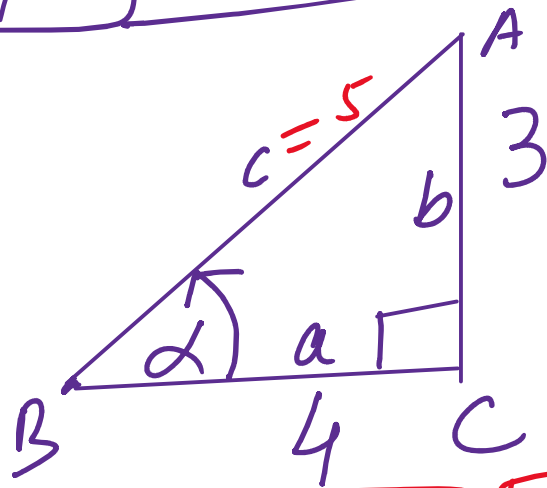
$$\boxed{m = \frac{-3}{1} = \tan \alpha}$$

$$|m = \frac{-2}{4}| = \tan \alpha$$

$$c^2 = a^2 + b^2$$

$$c = \sqrt{a^2 + b^2}$$

$$c = \sqrt{9 + 16} = 5$$

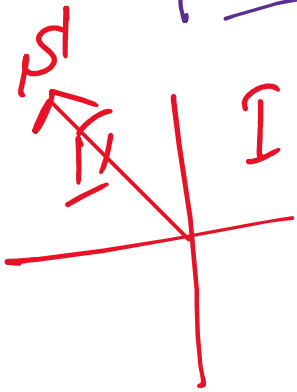


$$c = 5$$

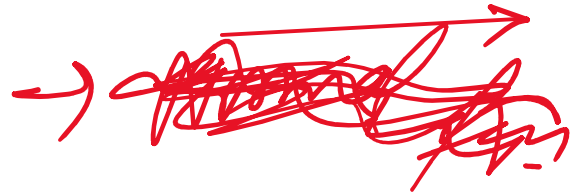
$$\sin \alpha = \frac{3}{5}$$

$$\cos \alpha = \frac{4}{5}$$

As α lies in the 2nd Quad.



$$3x + 4y - 12 = 0$$



On X-axis

X-intercept

$$y = 0, 3x = 12$$

$$x = 4$$

$$P(4, 0)$$

$$x_1, y_1$$

Thus (A) $\Rightarrow$

$$\boxed{\frac{x-4}{-\frac{4}{3}} = \frac{y-0}{\frac{3}{5}}}$$

This is the req. symmetric form

(a) (b) $\curvearrowright$

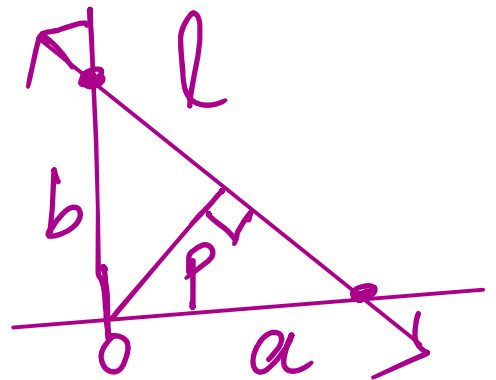
$$3x + 4y - 12 = 0$$

d) The req. Normal form is

$$\boxed{x \cos \alpha + y \sin \alpha = p}$$

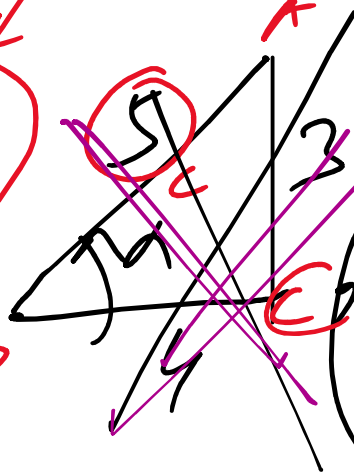
Sol.

$\sim$ Slope = $m = \tan \alpha = -\frac{3}{4}$



$$c = \sqrt{a^2 + b^2}$$

$$c = 5$$

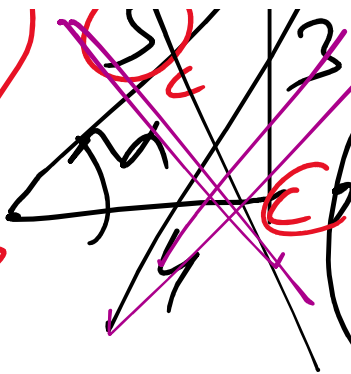


$$\cos \alpha = -\frac{4}{5}$$

$$\sin \alpha = \frac{3}{5}$$

$$c = \sqrt{a^2 + b^2}$$

$$c = 5$$



$$\cos \alpha = \frac{4}{5}$$

$$\sin \alpha = \frac{3}{5}$$

$$3x + 4y = 12$$

$\div 4y$ $\frac{3x}{4} + 1 = \frac{3}{4}$ on $\sin$, $\cos$

$$\frac{3x}{5} + \frac{4y}{5} = \frac{12}{5}$$

$$x\left(\frac{3}{5}\right) + y\left(\frac{4}{5}\right) = \left(\frac{12}{5}\right) \text{ req. 2}$$

$$\cos \alpha = \frac{3}{5}, \sin \alpha = \frac{4}{5}$$

(m, c)

$p = \frac{12}{5}$

Convert the equation $4x - 3y + 9 = 0$ into the given forms:

- Slope intercept
- Two intercepts
- Symmetric
- Normal

$$a) 3y = 4x + 9$$

$$\Rightarrow y = \frac{4}{3}x + 3$$

$$\begin{array}{c} -10 \quad -3 \\ \hline m = \frac{4}{3}, c = 3 \checkmark \end{array}$$

b) $4x - 3y = -9$

$$\Rightarrow \frac{4x}{-9} - \frac{3y}{-9} = 1$$

$$\frac{x}{\left(-\frac{9}{4}\right)} + \frac{y}{\left(\frac{9}{3}\right)} = 1$$

$$\Rightarrow \boxed{\frac{x}{\left(-\frac{9}{4}\right)} + \frac{y}{3} = 1}$$

x -intercept $= -\frac{9}{4}$ ✓
 y -intercept $= 3$ ✓

c) $4x - 3y + 9 = 0$

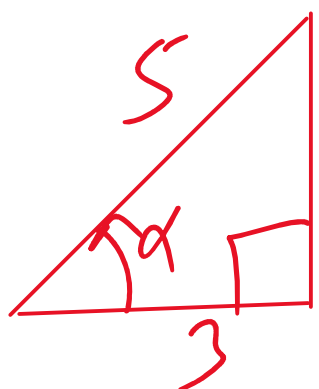
$$\frac{x-x_1}{\cos \alpha} = \frac{y-y_1}{\sin \alpha}$$

c) $4x - 3y + 9 = 0$

$\cos \alpha$ $\sin \alpha$

Symmetric for

slope $= m = \frac{-4}{-3} = \frac{4}{3} \checkmark$



$4 \Rightarrow \cos \alpha = \frac{3}{5}, \sin \alpha = \frac{4}{5}$

for point $A(x_1, y_1)$, on X-axis, $y = 0$
on Y-axis, $x = 0$ put in $4x = -9$
 $3y = 9 \Rightarrow y = 3$ $\Rightarrow$ $x = -\frac{9}{4}$ $A(-\frac{9}{4}, 0)$

Thus, req. symmetric for is

$$\frac{x - (-\frac{9}{4})}{\frac{3}{5}} = \frac{y - 0}{\frac{4}{5}}$$

or

$$\frac{x - 0}{2} = \frac{y - 3}{4}$$

$$\frac{x^2}{3/5} = \frac{y^2}{4/5}$$

d) Normal Form

$$4x - 3y + 9 = 0$$

$$x \cos \alpha + y \sin \alpha = p \quad \textcircled{1} \rightarrow \text{is regu.}$$

$$4x - 3y = -9$$

$\times$ by -1 on both sides

$$-4x + 3y = 9$$



As $\sqrt{4^2 + 3^2} = \sqrt{25} = 5$
 $\div$ by 5 on both sides, we get

$$-\frac{4x}{5} + \frac{3y}{5} = \frac{9}{5}$$

$$\Rightarrow \boxed{x \left(-\frac{4}{5} \right) + y \left(\frac{3}{5} \right) = \frac{9}{5}}$$

is the req. for

where $\cos \alpha = -\frac{4}{5}$, $\sin \alpha = \frac{3}{5}$

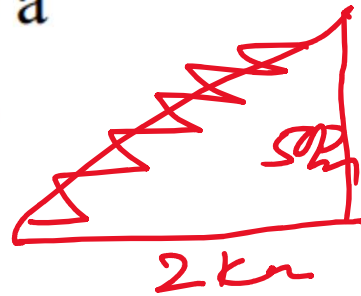
$$p = \frac{9}{5}$$

p = Perpendicular distance
of sight to the line

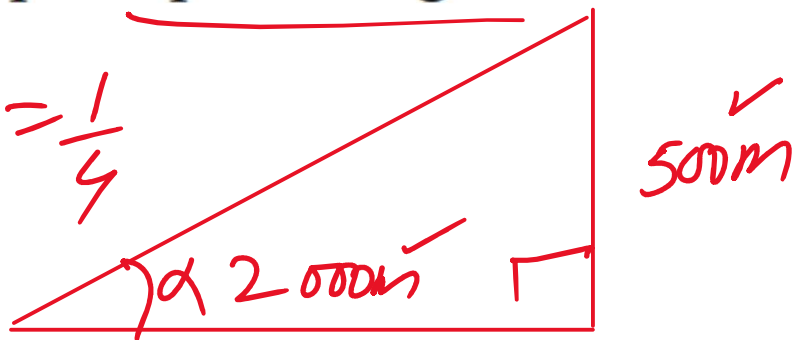
A hiking trail rises 500 meters over a horizontal distance of 2 kilometers.

What is the slope of a trail?

Express the slope in percentage.



$$m = \frac{500m}{2000m} = \frac{1}{4}$$



$$m = \frac{1}{4} \times 100\% = 25\%$$

$$m = \frac{1}{4} \times 100\% = 25\%$$

Transform this gen. str. line
 $ax + by + c = 0$ into slope-intercept

Sol $y = mx + c$

$$by = -ax - c$$

$$y = \left(-\frac{a}{b}\right)x + \left(-\frac{c}{b}\right)$$

is the req. form where
 $m = -\frac{a}{b}$, y-intercept $= -\frac{c}{b}$

2- $ax + by + c = 0 \rightarrow$ (1)
 point-slope form

The req. form is

$$y - y_1 = m(x - x_1) \rightarrow (A)$$

Slope of given line $= m = -\frac{a}{b}$

For a point on y-axis,
 put $x = 0$ put all

$$by = -c \Rightarrow y = -\frac{c}{b}$$

$$by = -c \Rightarrow y = -\frac{c}{b}$$

$$P(x_1, y_1) = P\left(0, -\frac{c}{b}\right)$$

Thus Eq. (1)

$$y - \left(-\frac{c}{b}\right) = -\frac{a}{b}(x - 0)$$

is the req. form

~~Two point form~~

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

(vi) Normal Form

The equation: $ax + by + c = 0$

(1)

can be written in the normal form as:

$$\frac{ax + by}{\pm \sqrt{a^2 + b^2}} = \frac{-c}{\pm \sqrt{a^2 + b^2}} \quad (2)$$

The sign of the radical to be such that the right hand side of (2) is positive.

Proof. We know that an equation of a line in normal form is

(3)

$$x \cos \alpha + y \sin \alpha = p$$

If (1) and (3) are identical, we must have

$$\text{i.e., } \frac{p}{-c} = \frac{\cos \alpha}{a} = \frac{\sin \alpha}{b} = \frac{\sqrt{\cos^2 \alpha + \sin^2 \alpha}}{\pm \sqrt{a^2 + b^2}} = \frac{1}{\pm \sqrt{a^2 + b^2}}$$

$$\text{Hence, } \cos \alpha = \frac{a}{\pm \sqrt{a^2 + b^2}} \quad \text{and} \quad \sin \alpha = \frac{b}{\pm \sqrt{a^2 + b^2}}$$

Substituting for $\cos \alpha$, $\sin \alpha$ and p into (3), we have

$$\frac{ax + by}{\pm \sqrt{a^2 + b^2}} = \frac{-c}{\pm \sqrt{a^2 + b^2}}$$

Thus (1) can be reduced to the form (3) by dividing it by $\pm \sqrt{a^2 + b^2}$. The sign of the radical to be chosen so that the right hand side of (2) is positive.